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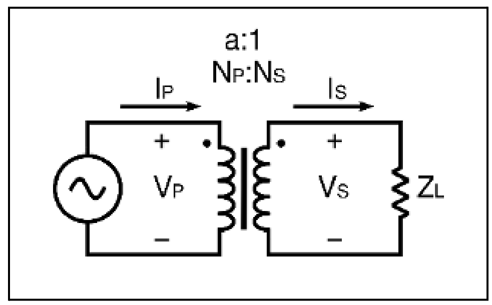
Transformers

Analysis

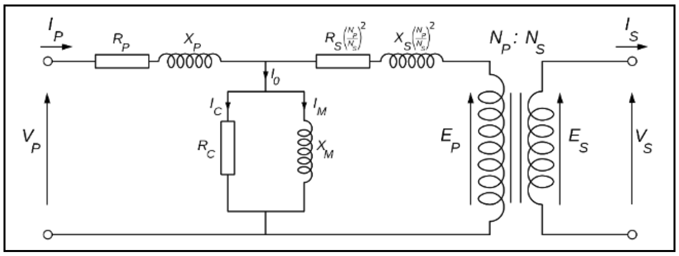
A transformer consists of two coils linked by a magnetic field. When voltage (VP) is applied to the first coil (the primary coil) an alternating current is created (IP), creating a magnetic field. This field induces a voltage (VS) in the secondary coil, causing a current to flow through a connected load (ZL)If the primary (NP) and secondary (NS) coils are on the same magnetic core (common for typical transformers), then the magnetic field is effectively the same. This results in a ratio relationship: $\frac{IS}{IP} = \frac{VP}{VS} = \frac{NP}{NS}$ where n stands for the number of coils.



In this lab, we assume that there is a perfect efficiency and that te power into the transformer equals the power out- essentially, $P_{primary} = P_{secondary}$. This means that $Vp\ *\ Ip\ = Vs\ *\ Is$. We can use this formula to solve unknown aspects about the circuit. This also shows that by increasing the voltage through a transformer we reduce the current, and since energy loss in a cable depends on the current and resistance of the wire, we therefore reduce energy loss.

Most devices are rated in watts or kilowatts (real power, P) , but transformers are rated in (kilo) volt amperes (apparent power, VA). As previously mentioned, the power going in on the primary side is the same as the power going out on the secondary side, and if secondary voltage is smaller then that means secondary current is increased so that the secondary power is equivalent to the primary power. So basically, transformers just transfer power between the coils, and the watts measurement depends on what you connect to the transformer. In AC circuits, the load depends on the true power (watts) and the power factor (the efficiency), and this all depends on the device, and the manufacturer doesn't know what people will connect to the transformer. Also, there's some wasted energy (reactive power) and we have to account for this as well, not just the true power, and we do so by measuring apparent power.

Transformers have four windings, and depending on how the circuit is wired we can get an output of 120v or 240v. Homes in the US use a three wire system, and the extra wire is connected to the center of the secondary coil, which means that we can use half of the coil or also all of the coil to get 240 volts The ratio $\frac{np}{ns}$ (the number of turns) is actually a limiting factor, so we can use parallel or series to create flexibility on our input and output. Depending on how the transformer is configured, the output secondary voltage can either be 120 or 240 volts. For higher voltage, we put the windings in series and each winding acts similar to a battery/ For higher current, we put the windings in parallel. The goal of this lab is to experience the effects of altering the arrangements of the wiring and see how that affects our data. By understanding how to work with and use transformers, we can use them in future circuits and experience all the benefits of transformers (such as electrical isolation).



Objectives

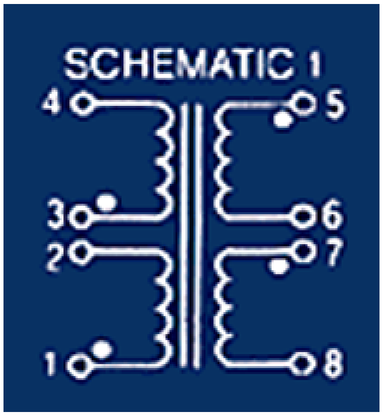
1. To experimentally and physically study a transformer, including its wiring, the role of the core Material, and the primary and secondary voltage and current relationships.

2. To understand typical multi-winding transformers wiring configurations and associated voltage regulation measurements.

3. To understand center-tap configurations and how this relates to the three-wire system used in residential 120/2400V electrical systems (3-wire Edison wiring configuration) including an examination of the voltage waveforms involved in such a system.

The wiring schematic of the transformer used in the lab is shown to the right. The data sheet for this transformer is as shown at:http://mcitransformer.com/product/mci-4-44-series/

Part 1:



IMG 1. Transformer schematic 1, used in part 1

a. First, an Ohmmeter was used to confirm the transformer coils as shown in the diagram of schematic one. All terminals (1 and 2, 3 and 4, 5 and 6, 7 and 8) were shown to have continuity.Vp was shown to be 49.1v, Vs was about 5.1v, and the transformer was approximately 10 to 1.

b. The resistance of all four winding connections were measured and recorded in table 1, as shown below.

R_{12}	R_{34}	R_{56}	R_{78}
80	90	1.9	1.8

Table 1. *The resistance of all four winding connections*

c. It was confirmed that there is no connections (and thus no resistance) between the separate windings R(1 or 2) and (3, 4, 5, 6, 7 or 8), R(3 or 4) and (5, 6, 7 or 8), R(5 or 6) and (7 or 8).

d. Based upon the above measurements and examining the actual windings (the relative wire sizes) we can infer about the relative number of winding turns that since $\frac{Vp}{Vs} = \frac{Is}{Ip} = \frac{Np}{Ns}$, $\frac{49.1}{5.1} = \frac{Is}{Ip} = \frac{Np}{Ns}$, and the primary current is therefore 1/10th of the secondary current. This checks out, because the primary winding has smaller wires, and thus more wires and more resistance, and thus more voltage and less current. Contrairly, the secondary current is larger and the secondary voltage is smaller.

$Vp = Ip * R$

$Vs = Is * R$

Where $\frac{49.1}{5.1} = \frac{Is}{Ip}$, $so\ 9.6 = \frac{Is}{Ip}$ $and\ Is = 9.6\ Ip$

$49.1/R = Ip$

$Is = (9.6)(49.1/R)$= 471.36/R

If similar windings have significantly different values, it could be due to a measurement error or unaccounted for leakage of inductance.

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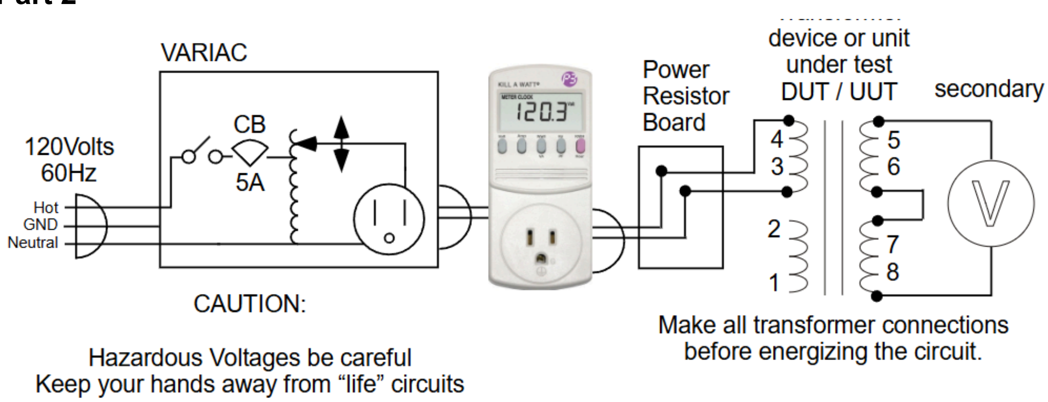
Part 2



a. Using the same circuit from schematic 1, the measured current is recorded in the table below. For comparison's sake, the calculated current is also in this table.

Primary current	**Secondary current**
0.03A	**3v/10 ohms=0.3A**

Table 2. *Calculated and measured current. The primary current was measured to be 0.03, and based on that the secondary current can be calculated to be 3v / 10 ohms, or 0.3 A. Since the current goes from 0,03A to 0,3A, it is a step up with a 1:10 ratio.*

b. Then, we altered the circuit by adding 10 ohms, measured the new primary and secondary voltage, and then we solved for the current. Since the load resistor is measured to be 10 ohms, we measured the output voltage and divided it by 10 to get the current.

Output current = $\frac{1.76v}{10\Omega} = 176mA$

Calculated input voltage $\frac{Vp}{Vs} = \frac{Is}{Ip}, \ \frac{v}{1.76v} = \frac{176mA}{19.2mA}\ , \ \frac{v}{176}\ = 9.17, \ vp = 9.17$.

Calculated output voltage $\frac{Vp}{Vs} = \frac{Is}{Ip}, \ \frac{17.6}{Vs} = \frac{176mA}{19.2mA}\ , \ \frac{v}{176}\ = 19.02, \ vs = 19.02$.

The ratio is more like 1:9.17 instead of 1:10, which leads to some differences between the measured and calculated data.

The efficieny is calculated as $\frac{power\ out}{power\ in} < \ 100\%, \ \frac{Vs\ Is}{Vp\ Ip} = \ \frac{19.02v * 176mA}{17{,}6\ * 19.2mA} = 0.72$

Input current	Input voltage	Calculated input voltage	Calculated input current	Output voltage	Output current	Calculated output voltage	Measured RL value	Efficiency
19.2 mA	17. 6V	9.17V	1.76mA	1.76 volts	176 mA	19.02V	10 ohms	0.72

Table 3. *Calculated and measured values for the altered circuit. The values were calculated using the formula* $\frac{Vp}{Vs} = \frac{Is}{Ip} = \frac{Np}{Ns}$.

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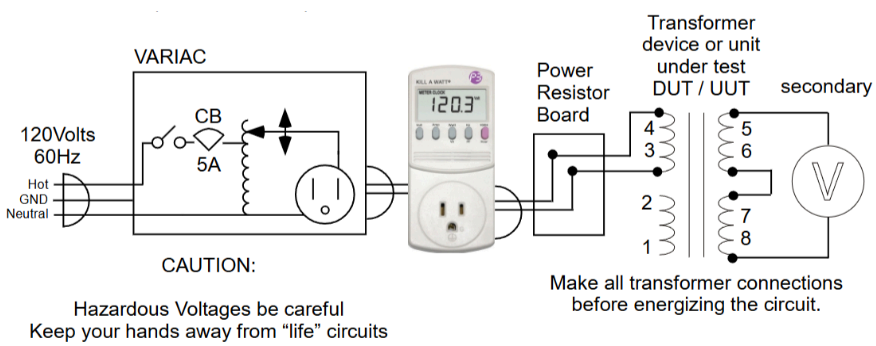
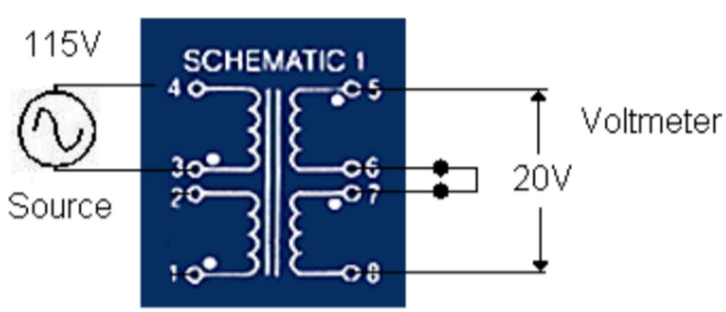
Part 3

We have already noticed the transformer used in this lab has two 115V windings and two 10V windings. This means that the transformer can be wired in several configurations! In the transformer diagrams below in each part of part three, the white dots on terminals 1,3,5 and 7 are polarity indicators- for example, when terminal 3 is positive with respect to terminal 4, then terminal 1 will be positive with respect to terminal 2. For series, the dot polarity should be positive to negative (or, “dot to no dot” . This is similar to a battery operation, and how when we want a higher voltage we connect plus and minus together in series, but if we want more current we want a parallel connection (plus to plus and minus to minus, dot to dot and no dot to no dot). For all of the 115v circuits in part three, the Kill A Watt meter is used, as shown in the example below:



3a-

The circuit for 3a was one primary and two secondaries connected in series.



IMG 2. *Schematic used in part 3a and 3b.*

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current
100	20	0	0	1.0	0

Table 4. *Information from 3a*

3b-

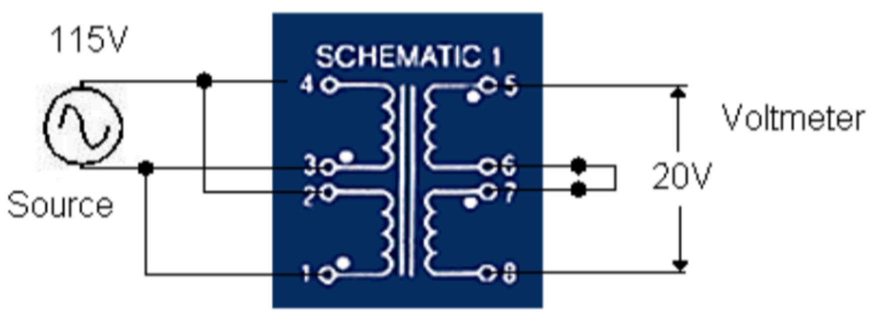
Part b is very similar to part a and uses a similar schematic, but a 21 ohm resistor is connected as the load.

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current	Load
100	16.5	23	23	0.98	0.24	21 ohms resistor

Table 5. *Information from 3b*

3c-

In part c, two primaries were interconnected in parallel, and two secondaries in series. The measurements for voltage wth a load and without a load (load of 21 ohms of resistance) was recorded (3ca without the load and 3cb with the load).



IMG 3. *Schematic used in part 3c*

3ca-

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current
99.8	20.6	0	0	1	0

Table 6. *Information from 3ca, circuit from part c without the load*

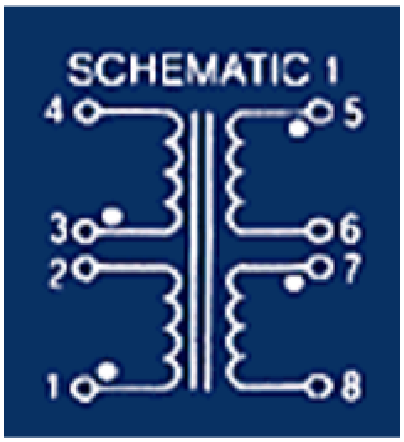
3cb-

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current	Load
99.4	16.4	23	23	0.97	0.24	21 ohms resistor

Table 7. *Information from 3cb, circuit from part c with the load*

3d-

In part 3d, two primaries are connected in parallel and two secondaries are connected in parallel.



IMG 4. *Schematic used in part 3d. The wires for this circuit are not shown in the schematic.*

3da-

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current	Load
99.7	10.2	0	0	1	0	21 ohms resistor

Table 8. *Information from 3da*

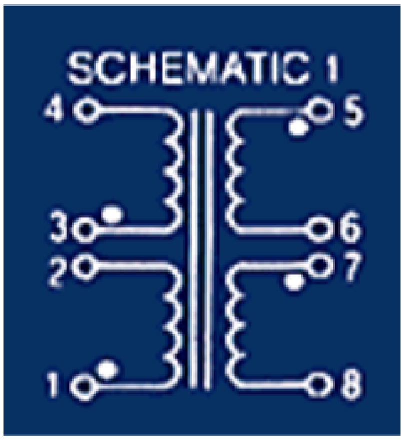
3db -

Input voltage	Output voltage	Power watts	Volt amps	Power factor	Input current	Load
99.8	9.6	0.8	10	0.85	0.10	21 ohms resistor

Table 9. *Information from 3db, circuit from part d with the load*

3e-

The schematic build in part 3e was two primaries connected in series, and two secondaries connected in parallel.



IMG 5. *Schematic used in part 3e The wires for this circuit are not shown in the schematic.*

3ea-

Input voltage	**Output voltage**
141.5	**14.53**

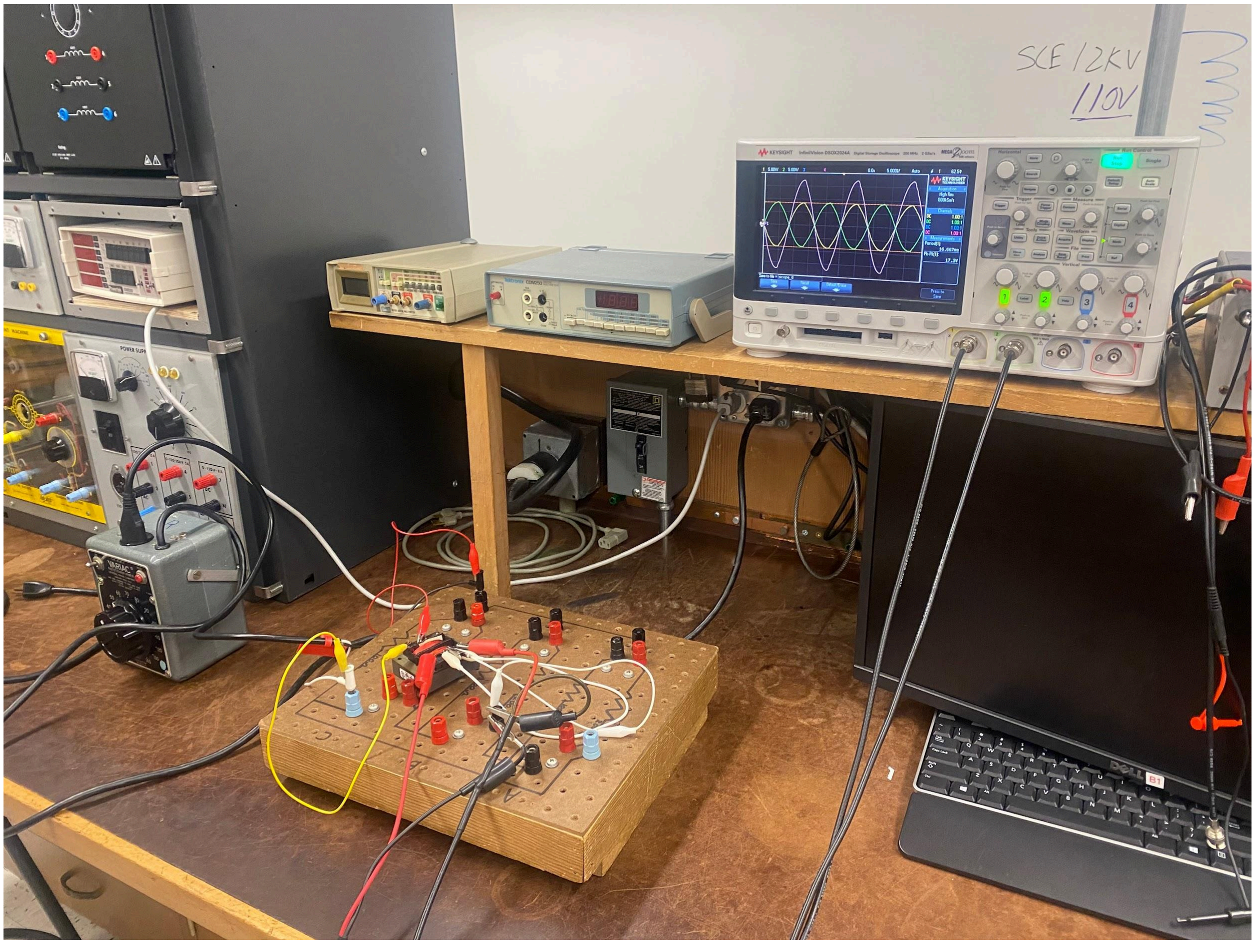
Table 10. *Information from 3ea*

3eb-

Input voltage	**Output voltage**	**Load**
141.1	**13.73**	**21 ohms resistor**

Table 11. *Information from 3eb, circuit from part e with the load*

Part 4- center tap



IMG 6. *Photo of center tap.*